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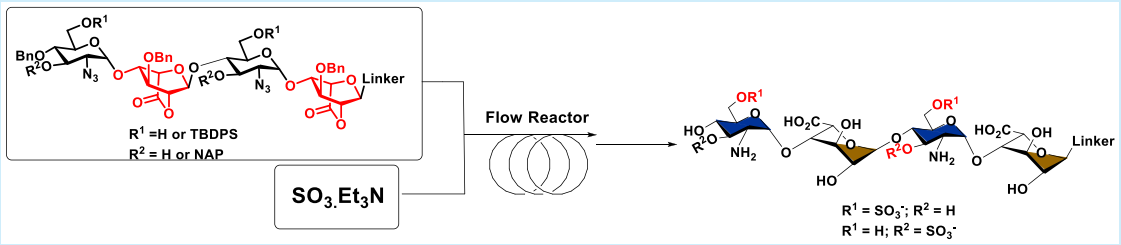
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Continuous-Flow Accelerated Sulfation of Heparan Sulfate Intermediates

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ABSTRACT: We report for the first time a continuous-flow strategy to execute *O*-sulfation modification of heparan sulfate (HS) oligosaccharides. A systematic investigation of the influence of the flow parameters on the installation of the sulfate group on glucosamine monosaccharide can aid the development of a comprehensive, quick, and reliable strategy for *O*-sulfation of HS oligosaccharide precursors. Deprotection of the sulfated heparin intermediates led to the development of a comprehensive biologically inspired oligosaccharide library to understand the crucial structure−function relationship of HS.

Heparan sulfate (HS) and heparin (HP) belong to the family of glycosaminoglycans (GAGs), which are linear polysaccharides that are ubiquitously distributed on all mammalian cell surfaces and mediate a wide range of biological processes.[1] One of the most striking features of HS and HP are their structural microheterogeneity, which originates from repeating disaccharide units (glucosamine and uronic acid), epimers of uronic acid (D-glucuronic and L-iduronic acid), and sulfate groups.[2] Among these modifications, the importance of the sulfate groups in HS- and HP-mediated interactions is well-recognized and has been studied extensively over the past two decades.[3] For example, the role of fibroblast growth factor (FGF) in fibroblast signaling depends on the 6-*O*-sulfated and *N*-sulfated glucosamine residues of HS,[4] and it has been shown that the 3-*O*-sulfated and *N*-sulfated glucosamine groups of fondaparinux and enoxaparin are important to regulate antithrombin III activity.[5] Previous research reported that 6-*O*-sulfated HS plays a pivotal role in heparin binding-epidermal growth factor (HB-EGF) and vascular endothelial growth factors (VEGF) binding affinity[6] and that 3-*O*-sulfation of HS is important for neuropilin-1 binding, tau protein binding, and herpes simplex virus 1 entry.[7] Because of the wide range of biological and pharmaceutical applications of the "sulfation code" of HS oligosaccharides, there have been increasing efforts to develop an efficient methodology for large-scale synthesis of well-defined fragments of HS oligosaccharides. Recently, several research groups have developed elegant strategies for stereoselective glycosylation, with careful selection of protecting groups to prepare the HS backbone at the commercial scale.[8] However, poor yields of poststructural modifications of the HS backbone, including oxidation and sulfation, as well as global deprotection have limited the large-scale synthesis of HS oligosaccharides.[9]

In the literature, *O*-sulfation of GAG intermediates is usually carried out using a sulfur trioxide complex (SO_3·NMe_3 or SO_3·NEt_3) by the batch method at 40−60 °C for 2−3 days.[9] However, the weakly nucleophilic hydroxyl groups of sugars and anionic repulsion between multiple sulfate groups of GAG oligosaccharides result in low yields and several partially sulfated side products. Therefore, gram-scale sulfation of oligosaccharides and selective multiple sulfation by such methods are still challenging. Alternatively, a chemoenzymatic method was employed to sulfate HS oligosaccharides (Scheme 1).[10] Although this method uses milder conditions, it requires expensive enzymes, a trained person, and a longer reaction time. Recently, microwave-mediated sulfation was developed to synthesize sulfated HS oligosaccharides at 100 °C within 15−20 min.[11] However, this method is restricted to a specific substrate scope and forms decomposition side products, which restricted scale-up of this method.[12] In contrast, flow chemistry

Received: March 9, 2020
Published: April 20, 2020

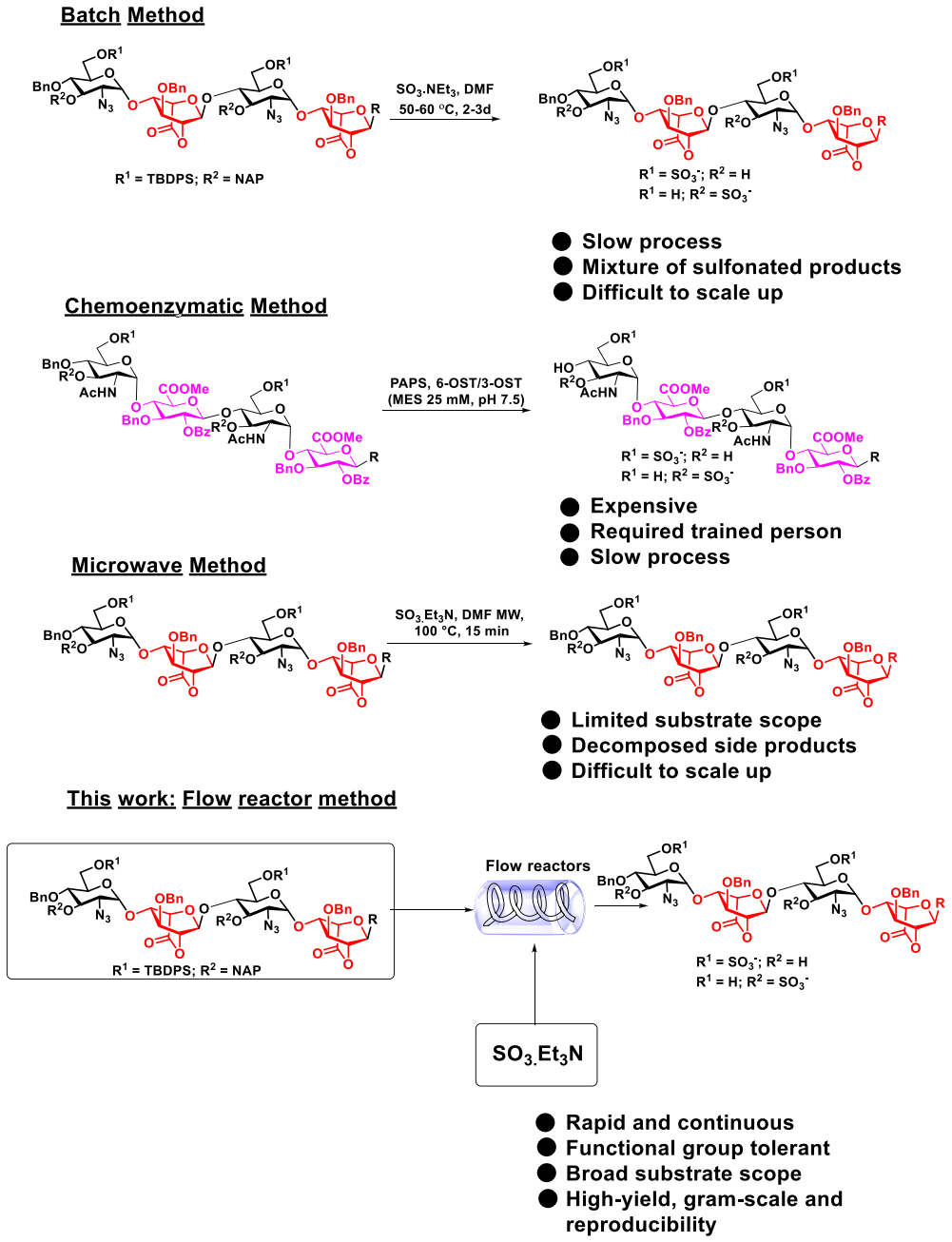


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Org. Lett. 2020, 22, 3402−3406

Scheme 1. Sulfation of HS Oligosaccharides by Prior Methods



has several milestone achievements in chemical processes and transformations. Furthermore, flow chemistry troubleshoots the difficulties faced in the batch process, leading to fascinating results, including improved slow reaction kinetics by precise control over the reaction conditions, including temperature, pressure, and uniform mixing of the reactants.[13] On the basis of such advantages, herein we studied the mono- and multiple sulfation of biologically inspired oligosaccharides to synthesize sulfated HS oligosaccharide intermediates for the first time. Furthermore, concomitant deprotection of the final sulfated oligosaccharides resulted the desired HS oligosaccharides. The novel sulfation approach described herein is general and could be applied to other GAG oligosaccharides, such as dermatan, chondroitin, and keratan sulfates.

In the quest to develop a mild, selective, and high-yielding *O*-sulfation reaction in a continuous flow system, we first standardized the flow optimization on 6-*O*- and 3-*O*-sulfation of a glucosamine precursor. Glucosamine substrate **1** was converted to sulfate precursor **2** by deprotection of the TBDPS group using HF·pyridine. A 0.05 M solution of **2** in DMF and a 0.5 M solution of SO_3·NEt_3 in DMF were pumped at 0.05 and 0.1 mL/min, respectively, into a 10 mL stainless steel coiled reactor in a Vaportec R-series system at room temperature and 3.5 bar pressure, giving a retention time of 66.6 min (graph (i) in Figure 1). Evaporation of the volatile solvent from the collected fractions followed by column chromatography purification afforded the product **3** in 68% yield (Table S1). Increasing the flow rate (0.1 mL/min for **2** and 0.2 mL/min for SO_3·NEt_3, retention time = 33.3 min) resulted in a slight decrease in yield to 62% (graph (ii) in Figure 1). Byproducts formed by deprotection of Cbz/NAP/Troc groups were not observed under continuous flow. Using higher temperatures

https://dx.doi.org/10.1021/acs.orglett.0c00878
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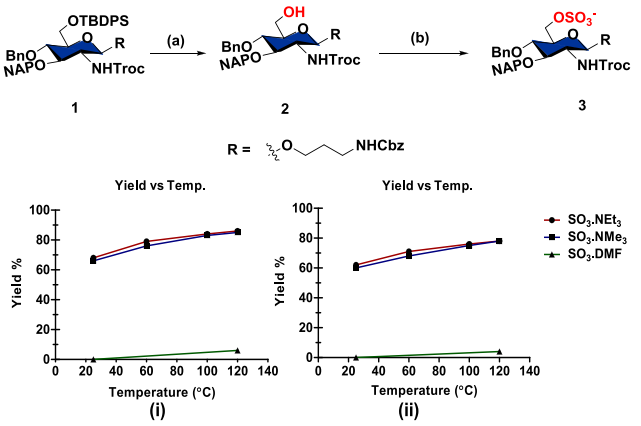


Figure 1. (top) Continuous 6-*O*-sulfation of glucosamine precursor **2**. Reagents and conditions: (a) 70% HF·py, py, 0 °C, 12 h; (b) SO_3·NEt_3 or SO_3·NMe_3 or SO_3·DMF in DMF and **2** in DMF were flowed into the 10 mL stainless steel coil flow reactor (Vaportec R-series). (bottom) Comparisons of 6-*O*-sulfation yields vs temperature for different sulfur trioxide precursors at 3.5 bar pressure: (i) retention time = 66.6 min (**2** in DMF at 0.05 mL/min and SO_3·NEt_3 in DMF at 0.1 mL/min); (ii) retention time = 33.3 min (**2** in DMF at 0.1 mL/min and SO_3·NEt_3 in DMF at 0.2 mL/min).

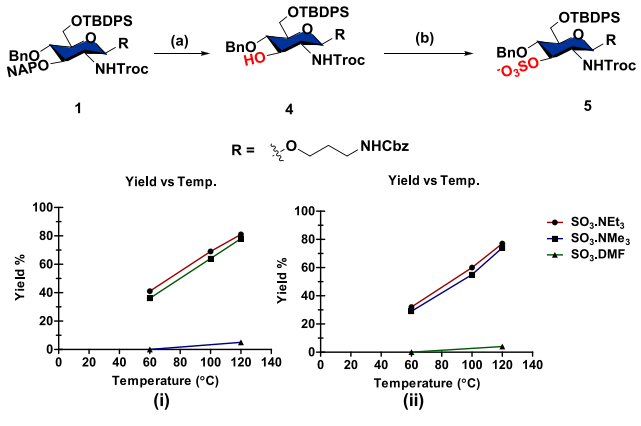


Figure 2. (top) Continuous 3-*O*-sulfation of glucosamine precursor **4**. Reagents and conditions: (a) DDQ, DCM/H_2O (18:1), RT, 1 h; (b) SO_3·NEt_3 or SO_3·NMe_3 or SO_3·DMF. (bottom) Comparisons of 3-*O*-sulfation yields vs temperature for different sulfur trioxide precursors at 3.5 bar pressure: (i) retention time = 66.6 min (**4** in DMF at 0.05 mL/min and SO_3·NEt_3 in DMF at 0.1 mL/min), pressure; (ii) retention time = 33.3 min (**4** in DMF at 0.1 mL/min and SO_3·Et_3N in DMF at 0.2 mL/min).

(60−120 °C) with the same retention time (66.6 min) and pressure (3.5 bar) resulted in moderate to good yields of **3** (71−86%).

Under the same conditions, SO_3·NMe_3 as the precursor gave similar yields (Table S2), indicating that both precursors were ideal for the *O*-sulfation reaction. However, the use of SO_3·DMF resulted in a 6% yield. A control experiment was performed under batch conditions and gave a 73% yield of **3** after 24 h. Overall, we optimized the retention time, temperature, and pressure essential for effective 6-*O*-sulfation of glucosamine **1** and showed that the sulfation of the glucosamine precursor is more efficient in the flow reactor compared with the batch process

Having established the optimum flow conditions for 6-*O*-sulfated glucosamine **3**, we next sought to investigate 3-*O*-sulfation of glucosamine **4** under optimized continuous-flow reaction conditions. Because of the steric hindrance and poor nucleophilicity of the 3-hydroxyl group of glucosamine **4**, we observed the sulfation reaction at C-3 to be slow (graphs (i) and (ii) in Figure 2). At room temperature, no conversion was observed at flow rates of 0.05 mL/min for glucosamine precursor **4** and 0.1 mL/min for the SO_3·NEt_3 precursor. As the temperature increased, the conversion rate showed a nearly linear correlation with the yield. Maximum conversion of **4** was observed at 120 °C with the SO_3·NEt_3 complex. With an increase in the precursor flow rate from 0.1 to 0.2 mL/min, the yield of the sulfation is decreased (Tables S3 and S4). This result suggests that the flow rate of the precursor exerts a strong influence on 3-*O*-sulfation. A similar reaction in a batch process afforded a 65% yield of **5** after 2 days at 60 °C, and at a high temperature (120 °C) we observed fragmentations. Overall, these results revealed that a fine optimization between the retention time of the precursor and a high reaction temperature was essential for 3-*O*-sulfation of glucosamine in the flow reactor.

After having optimized the flow conditions for *O*-sulfation using glucosamine residues, we were more interested in studying the continuous-flow *O*-sulfation chemistry with more fascinating biomolecules, heparinoids. Surprisingly, the flow chemistry provided excellent sulfation with di- and tetrasaccharide intermediates with different sulfation patterns in better yields than for the batch process (Figure 3). Disaccharide **6a**

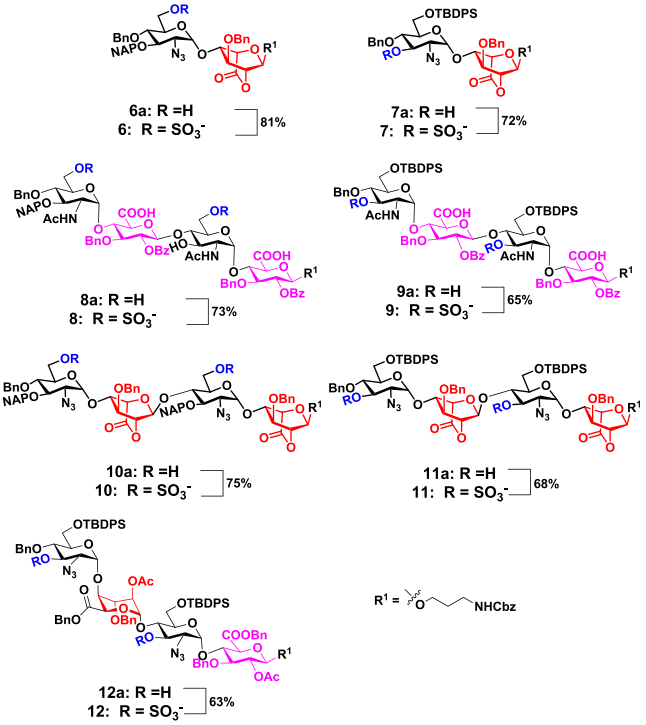


Figure 3. Substrate scope of mono- and disulfation of disaccharides and tetrasaccharides.

was persulfated at the 6-position using an optimized flow rate in an excellent yield of 81%. Similarly, 3-*O*-sulfation of disaccharide **7** furnished 72% conversion. In all cases, a minor quantity of unreacted starting material was observed in the reaction mixture and recovered by column purification for further reaction. Tetrasaccharides **8a−12a** with different

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uronic acid compositions (L-iduronic and D-glucuronic acid precursors) and glucosamine derivatives showed excellent yields of 6-*O*- and 3-*O*-sulfation products (63−75%). Moreover, the gram-scale conversion of **6a** demonstrated the potential of the flow chemistry. Finally, persulfated **6** and **8**−**11** were then fully deprotected using HF·pyridine (in the case of **9** and **11**) followed by LiOH·H_2O and hydrogenolysis to afford the desired HS tetrasaccharides **13**−**17** (Figure 4).

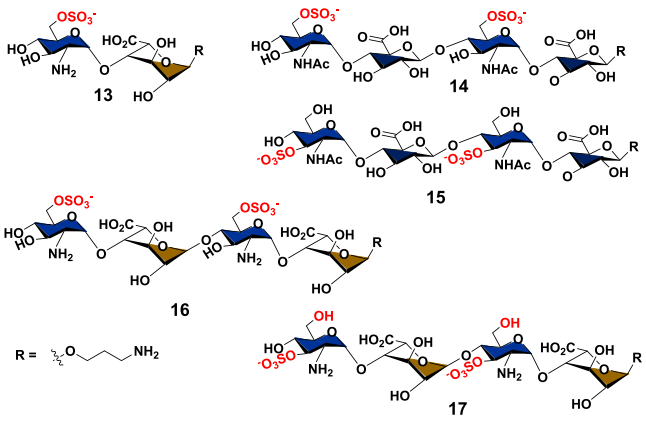


Figure 4. Heparan sulfate oligosaccharides **13**−**17** after complete deprotection.

In conclusion, machine-accelerated *O*-sulfation of biomolecules such as HS oligosaccharides in a continuous flow system was developed for the first time. At an optimized flow rate, temperature, and pressure, the 6-*O* and 3-*O* hydroxyl groups of glucosamine, HS disaccharide, and HS tetrasccharide intermediates were effectively sulfated in excellent yields (65−81%). The resulting tetrasaccharides were fully deprotected under mild conditions to yield the desired HS oligosaccharides (>88%). Using this approach, we synthesized different uronic acids composed of tetrasaccharides. The generality of the reaction conditions and easy to scale-up process can be expected to accelerate the synthesis of HS oligosaccharides with different sulfation patterns. We are currently investigating simultaneous 6-*O*/3-*O*-sulfation and *N*-sulfation.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.0c00878.

Experimental procedures and spectroscopic characterization (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

Financial support from IISER-Pune, DST (Grant SR/NM/NS-1113/2016), and DBT (Grant BT/PR21934/NNT/28/1242/2017) is gratefully acknowledged.

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